

Last time:

• $C(X) = \{f: X \rightarrow \mathbb{R} : f \text{ continuous and bounded}\}$
is a metric space

• Prop: $f_n \rightarrow f$ in $C(X) \iff f_n$ converges uniformly to f

• Thm: $\left. \begin{array}{l} f_n \text{ converges uniformly to } f \\ f_n \text{ continuous } \forall n \end{array} \right\} \Rightarrow f \text{ continuous}$

Thm Let X be a metric space. Then the metric space $(C(X), d_{\text{sup}})$ is complete.

Pf: Fix a Cauchy sequence $\{f_n\}$ in $C(X)$.

Last time:

• $\exists f: X \rightarrow \mathbb{R}$ s.t. $f_n(x) \rightarrow f(x) \quad \forall x \in X$

• Claim: f_n converges uniformly to f .

Altogether:

① f is continuous: by the previous theorem + claim

② f is bounded: By the claim, $\exists N$ s.t.

$$|f_N(x) - f(x)| < 1 \quad \forall x \in X$$

$$\underline{f_N \in C(X)} \Rightarrow f_N \text{ bounded}$$

$$\Rightarrow \exists M > 0 \text{ s.t. } |f_N(x)| \leq M \quad \forall x$$

$$\Rightarrow |f(x)| \leq |f(x) - f_N(x)| + |f_N(x)| \leq 1 + M$$

③ $f \in C(X)$: by ① + ②

④ $f_n \rightarrow f$ in $(C(X), d_{\text{sup}})$: by the previous proposition + claim. $\square$

Thm (Cauchy criterion) Let X be a metric space and $f_n: X \rightarrow \mathbb{R} \quad \forall n \geq 1$. Then: $\{f_n\}_{n \geq 1}$ converges uniformly $\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t.

$$|f_n(x) - f_m(x)| < \varepsilon \quad \forall m, n \geq N \text{ and } \forall x \in X.$$

Pf: $\Leftarrow$: By the claim in the previous proof.

$\Rightarrow$: Suppose $\{f_n\}$ converges uniformly to f .

Fix $\varepsilon > 0$. Then $\exists N$ s.t.

$$|f_n(x) - f(x)| < \frac{\varepsilon}{2} \quad \forall n \geq N \text{ and } \forall x \in X.$$

So for $m, n \geq N$ we have

$$\begin{aligned} |f_n(x) - f_m(x)| &\leq |f_n(x) - f(x)| + |f(x) - f_m(x)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

for any $x \in X$. □

Def Let X be a metric space and $f_n: X \rightarrow \mathbb{R} \quad \forall n \geq 1$.

We say that the series of functions $\sum_{n=1}^{\infty} f_n(x)$

converges pointwise/uniformly to $f: X \rightarrow \mathbb{R}$ if the

sequence of functions $s_n = f_1 + \dots + f_n$ converges pointwise/uniformly to f .

Thm (Weierstrass M-test) Let X be a metric space and $f_n: X \rightarrow \mathbb{R} \quad \forall n \geq 1$. Suppose that $\forall n \geq 1, \exists$ a constant $M_n \geq 0$ s.t.

$$|f_n(x)| \leq M_n \quad \forall x \in X.$$

If $\sum_{n=1}^{\infty} M_n$ converges, then $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly.

Pf: We'll use the Cauchy criterion. Fix $\varepsilon > 0$.

As $\sum M_n$ converges, $\exists N$ s.t.

$$\sum_{k=m}^n M_k < \varepsilon \quad \forall n \geq m \geq N.$$

Set $s_n(x) = \sum_{k=1}^n f_k(x)$. Then, for $n > m \geq N$ we have

$$\begin{aligned} |s_n(x) - s_m(x)| &= \left| \sum_{k=m+1}^n f_k(x) \right| \leq \sum_{k=m+1}^n |f_k(x)| \\ &\leq \sum_{k=m+1}^n M_k < \varepsilon \quad \forall x \in X. \end{aligned}$$

For $m > n$ we can swap m and n above, and when $m = n$ we have $|s_n(x) - s_m(x)| = 0 < \varepsilon$.

Altogether, we conclude

$$|s_n(x) - s_m(x)| < \varepsilon \quad \forall m, n \geq N \text{ and } \forall x \in X.$$

So $\{s_n\}$ converges uniformly by the Cauchy criterion. $\square$

Ex Let $\{c_n\}_{n \geq 0}$ be a sequence of real numbers.

For which $x \in \mathbb{R}$ does the power series $\sum_{n=0}^{\infty} c_n x^n$ converge?

Fix $r > 0$. For $|x| \leq r$, we have

$$|c_n x^n| = |c_n| \cdot |x|^n \leq |c_n| \cdot r^n \quad \forall n \geq 0.$$

If $\sum |c_n| r^n$ converges, then by the Weierstrass M-test $\sum c_n x^n$ converges uniformly. Let's apply the root test to $\sum |c_n| r^n$:

$$1 > \limsup_{n \rightarrow \infty} (|c_n| r^n)^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} (|c_n|^{\frac{1}{n}} \cdot r) \\ = r \cdot \limsup_{n \rightarrow \infty} |c_n|^{\frac{1}{n}}$$

$$\Leftrightarrow r < \underbrace{\frac{1}{\limsup_{n \rightarrow \infty} |c_n|^{\frac{1}{n}}}}_{\text{Set } = R}$$

Conclude: For $r < R$, $\sum |c_n| r^n$ converges by the root test, and so $\sum c_n x^n$ converges uniformly for $x \in [-r, r]$ by the Weierstrass M-test.

Thm Let $\{c_n\}_{n \geq 0}$ be a sequence of real numbers and consider the power series $\sum_{n=0}^{\infty} c_n x^n$. Define the radius of convergence

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |c_n|^{\frac{1}{n}}} \in [0, \infty]$$

(with the convention $\frac{1}{+\infty} = 0$ and $\frac{1}{0} = +\infty$). Then:

① $\forall x \in (-R, R)$ the series $\sum c_n x^n$ converges, and the function $f(x) = \sum c_n x^n$ is continuous on $(-R, R)$.

② $\forall |x| > R$ the series $\sum c_n x^n$ diverges.

Rmk • For $x = \pm R$ the theorem makes no claim, and the power series may converge or diverge.

• By Cesaro-Stolz, if $\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$ exists then $\limsup_{n \rightarrow \infty} |c_n|^{1/n} = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$. This gives us an easier way to compute R for some examples.

Pf: ① We already showed that $\forall r \in (0, R)$, $\sum c_n x^n$ converges uniformly to some $f(x)$ for $x \in [-r, r]$. As each partial sum $S_N = \sum_{n=0}^N c_n x^n$ is continuous, then f is continuous on $[-r, r]$ (by lecture 22) $\forall r \in (0, R)$.

② Fix $r > R$. Then

$$r > R = \frac{1}{\limsup |c_n|^{1/n}} \Rightarrow \limsup |c_n|^{1/n} > \frac{1}{r}$$

$$\Rightarrow |c_n|^{1/n} > \frac{1}{r} \text{ for infinitely many } n$$

$$\Rightarrow |c_n| r^n > 1 \text{ for infinitely many } n.$$

So, for $|x| \geq r$,

$$|c_n x^n| = |c_n| \cdot |x|^n \geq |c_n| r^n > 1 \text{ for infinitely many } n$$

$$\Rightarrow c_n x^n \not\rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow \sum c_n x^n \text{ diverges.}$$

□